



Cambridge International A Level

MATHEMATICS

9709/33

Paper 3 Pure Mathematics 3

May/June 2026

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **27** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
	Ignore subsequent work
	Judgement
	Judgement
	Method mark awarded zero
	Method mark awarded one
	Method mark awarded two
	Misread
	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Judgment made by the PE
	Premature approximation
	Special case
	Indicates that work/page has been seen

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Annotation	Meaning
	Error in number of significant figures
	Correct
	Transcription error
	Correct answer from incorrect working

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

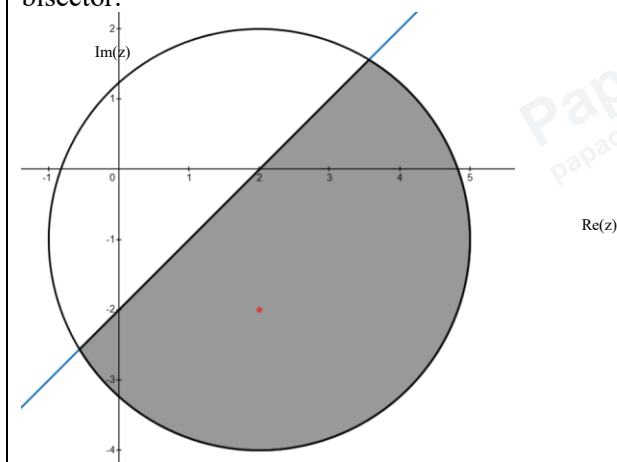
AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

Question	Answer	Marks	Guidance
1	Use correct law of logarithm for a power or for a sum	*M1	Correct use of brackets SOI by later work.
	Remove logarithms	DM1	E.g. $5(3x - 2) = (x - 4)^2$. Condone use of $\frac{\ln(x - 4)^2}{\ln(3x - 2)}$ but ‘recovered’ later. Must see evidence of recovery
	Obtain a correct quadratic equation, e.g. $x^2 - 23x + 26 = 0$	A1	Or three-term equivalent.
	Obtain final answer $x = \frac{23 + 5\sqrt{17}}{2}$ only	A1	Or exact equivalent.
		4	

Question	Answer	Marks	Guidance
2	EITHER: State or imply non-modular inequalities $3^{2x-1} - 4 < -2$ and $3^{2x-1} - 4 > 2$, or $(3^{2x-1} - 4)^2 > 2^2$ or corresponding pair of linear equations or quadratic equation	(B1	
	Use correct method for solving an equation of the form $3^{2x-1} = a$ where $a > 0$	M1	
	Obtain critical values $x = 0.815$ and 1.32 or exact equivalents	A1	E.g. $\left(\frac{\ln 6}{\ln 3} + 1\right) \div 2$ and $\left(\frac{\ln 2}{\ln 3} + 1\right) \div 2$. Allow greater accuracy, 1.31546..., 0.8154648...
	State final answer $x < 0.815, x > 1.32$	A1)	
	OR:		
	Use correct method for solving an equation of the form $3^{2x-1} = a$ where $a > 0$	(M1	
	Obtain one critical value, e.g. 0.815 or exact equivalent	A1	E.g. $\left(\frac{\ln 6}{\ln 3} + 1\right) \div 2$. Allow greater accuracy, 0.8154648....
	Obtain the other critical value, e.g. 1.32 or exact equivalent	A1	E.g. $\left(\frac{\ln 2}{\ln 3} + 1\right) \div 2$. Allow greater accuracy, 1.31546....
	State final answer $x < 0.815, x > 1.32$	A1)	

Question	Answer	Marks	Guidance
2	OR:		
	Use $y = 3^{2x}$ and hence obtain $\left(\frac{1}{3}y - 4\right)^2 > 4$	(B1)	OE
	Use correct method for solving an equation of the form $3^{2x} = a$ where $a > 0$	M1	
	Obtain critical values $x = 0.815$ and 1.32 or exact equivalents	A1	E.g. $\left(\frac{\ln 6}{\ln 3} + 1\right) \div 2$ and $\left(\frac{\ln 2}{\ln 3} + 1\right) \div 2$. Allow greater accuracy, 1.31546..., 0.8154648....
	State final answer $x < 0.815, x > 1.32$	A1)	Mark final answer. Inclusion of 'and' is A0.
	Available marks	4	

Question	Answer	Marks	Guidance
3	Show a circle centre $(2, -1)$	B1	Allow if stated but must have a circle shown with no contradiction.
	Show a circle with radius 3	B1FT	FT centre not at the origin. Allow if stated but must have a circle shown with no contradiction. Must not pass through $(5, 0)$ and $(-1, 0)$.
	Show the point representing $(2, -2)$ or the midpoint $(1, -1)$ or the straight line passing through the origin and $(2, -2)$ or the midpoint $(1, -1)$ Or use $y = x - 2$ (SOI by diagram)	B1	
	Show the perpendicular bisector of the line joining $(2, -2)$ and the origin, passing through or near $(0, -2)$ and $(2, 0)$ Or show the line $y = x - 2$ passing through $(0, -2)$ and $(2, 0)$	B1FT	FT is on the position of $(2, -2)$.
	Shade the correct region	B1	Dependent on all previous marks. SC allow B1 after B0 for inaccurate perpendicular bisector.
		5	



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Question	Answer	Marks	Guidance
4	Use correct trig formulae for $\cos(2\theta - 45^\circ)$ or for $\sin(2\theta - 45^\circ)$	B1	E.g. $\cos 2\theta \cos 45 + \sin 2\theta \sin 45$ or $\sin 2\theta \cos 45 - \cos 2\theta \sin 45$. Negative angles must be simplified correctly.
	Use correct trig formulae and substitute values for $\cos 45$ and $\sin 45$	*M1	
	Obtain correct simplified equation, e.g. $\sqrt{2} \sin 2\theta = -1$	A1	OE
	Solve <i>their</i> equation in $\sin 2\theta$ correctly to obtain a value for θ	DM1	
	Obtain answer, e.g. -22.5	A1	
	Obtain all four answers -67.5° , -22.5° , 112.5° and 157.5° and no others in the interval	A1	Treat answers in degrees as a misread. Withhold one A1 only for any instances of this throughout. Correct radian answers -1.179 , -0.393 , 1.96 , 2.75 Ignore answers outside the given interval.
		6	

Question	Answer	Marks	Guidance
5(a)	Obtain $\frac{dx}{dt} = 2\sec^2 2t$	B1	OE
	Attempt chain rule on $(\sin 2t)^{-1}$ for $\frac{dy}{dt}$, or attempt to use given formula	M1	I.e. $\frac{dy}{dt} = -(\sin 2t)^{-2} \frac{d}{dt}(\sin 2t)$. Use of quotient rule must give the form $p(\sin 2t)^{-2} \cos 2t$ OE. Use of formula $q \operatorname{cosec} 2t \cot 2t$.
	Obtain $\frac{dy}{dt} = -2(\sin 2t)^{-2} \cos 2t$	A1	OE
	Use $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$	M1	Expect $-2(\sin 2t)^{-2} \cos 2t \times \frac{\cos^2 2t}{2}$ OE if correct. Use <i>their</i> $\frac{dy}{dt}$ and $\frac{dx}{dt}$ correctly. Condone errors when dealing with the 2's.
	Obtain $\frac{dy}{dx} = \frac{-\cos 2t}{\tan^2 2t}$	A1	WWW Allow from $-\frac{\cos^3 2t}{\sin^2 2t}$. Must be in this form of answer given in the question.
		5	

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Question	Answer	Marks	Guidance
5(b)	Obtain $x = 1$ and $y = \sqrt{2}$ when $t = \frac{1}{8}\pi$	B1	
	Substitute $t = \frac{1}{8}\pi$ into <i>their</i> $\frac{dy}{dx}$ of the correct form	*M1	Expect $\frac{dy}{dx} = -\frac{\sqrt{2}}{2}$ if correct, $\frac{k}{\sqrt{2}}$ if incorrect.
	Form equation of the tangent with <i>their</i> (x, y) and <i>their</i> $\frac{dy}{dx}$	DM1	E.g. $y - \sqrt{2} = -\frac{\sqrt{2}}{2}(x - 1)$ if correct.
	Obtain equation of tangent $\sqrt{2}y + x = 3$ or exact three-term equivalent	A1	E.g. $2y = -\sqrt{2}x + 3\sqrt{2}$.
		4	

Question	Answer	Marks	Guidance
6	$\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$ or $\frac{dx}{du} = 2u$	B1	SOI
	Reach an integral of the form $\int A \cos 3u \, du$	*M1	OE Condone loss of du .
	Obtain $\int 2 \cos 3u \, du$	A1	OE
	Obtain $\frac{2}{3} \sin 3u$	A1FT	OE FT <i>their</i> coefficient A .
	Substitute correct limits correctly in an expression of the form $B \sin 3u$ or $B \sin 3\sqrt{x}$	DM1	$u = \frac{1}{2}\pi$ and $u = \frac{1}{4}\pi$, $x = \frac{1}{4}\pi^2$ and $x = \frac{1}{16}\pi^2$.
	Obtain answer $-\frac{2}{3} - \frac{1}{3}\sqrt{2}$	A1	Or exact two-term equivalent. Accept, e.g., $-\frac{2}{3}\left(1 + \frac{1}{\sqrt{2}}\right)$. ISW
		6	

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Question	Answer	Marks	Guidance
7	Square $x + iy$ and equate real and imaginary parts to 3 and $-20\sqrt{7}$ respectively	*M1	Allow use of $a + bi$ or similar throughout.
	Obtain equations $x^2 - y^2 = 3$ and $2xy = -20\sqrt{7}$	A1	OE
	Eliminate one variable correctly and find an equation in the other	DM1	Attempt to find $x^2 - y^2 [= 3]$ and $(x^2 + y^2)^2 [= 53^2]$ instead.
	Obtain $x^4 - 3x^2 - 700 = 0$ or $y^4 + 3y^2 - 700 = 0$, or three-term equivalents	A1	Allow $x^2 - y^2 = 3$ and $x^2 + y^2 = 53$ instead.
	Obtain answers $\pm(2\sqrt{7} - 5i)$	A1	Accept $\sqrt{28}$.
		5	

Question	Answer	Marks	Guidance
8(a)	Sketch $y = x^3$ and $y = 4 - 2x$ and justify the given statement	B1	Sketch must show all four quadrants. May also sketch $y = x^3 + 2x$ and $y = 4$, $y = x^3 - 4$ and $y = -2x$, or $y = 4 - x^3$ and $y = 2x$. Need to see a statement about one point of intersection, or a cross, a circle or a dotted line.
		1	

Question	Answer	Marks	Guidance
8(b)	EITHER: Express $x_{n+1} = \frac{3x_n^3 + 4}{4x_n^2 + 2}$ as $x = \frac{3x^3 + 4}{4x^2 + 2}$	(M1)	
	Rearrange $x = \frac{3x^3 + 4}{4x^2 + 2}$ to $x^3 = 4 - 2x$ with full and correct working	A1)	AG
	OR:		
	Rearrange $x^3 = 4 - 2x$ to $x = \frac{3x^3 + 4}{4x^2 + 2}$	(M1)	
	Express $x = \frac{3x^3 + 4}{4x^2 + 2}$ as $x_{n+1} = \frac{3x_n^3 + 4}{4x_n^2 + 2}$ after full and correct working	A1)	AG
	Available marks	2	

Question	Answer	Marks	Guidance
8(c)	Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 2$	M1	$f(x) = x^3 - 4 + 2x$: $f(1) = -1 < 0$ $f(2) = 8 > 0$. Or comparing x^3 and $4 - 2x$: $x^3 = 1$ and $4 - 2x = 2$, $1 < 2$ $x^3 = 8$ and $4 - 2x = 0$, $8 > 0$. Or $f(x) = \frac{3x^3 + 4}{4x^2 + 2} - x$: $f(1) = 0.16... > 0$ $f(2) = -0.44... < 0$. Or comparing $\frac{3x^3 + 4}{4x^2 + 2}$ and x : $\frac{3x^3 + 4}{4x^2 + 2} = 1.16... \quad x = 1, 1.16... > 1$ $\frac{3x^3 + 4}{4x^2 + 2} = 1.55... \quad x = 2, 1.55... < 2$. Allow if a smaller range is used provided there is a sign change within that range
	Complete the argument correctly with correct calculated values	A1	
		2	

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Question	Answer	Marks	Guidance
8(d)	Use the iterative formula correctly at least twice	M1	
	Obtain final answer 1.18	A1	
	Show sufficient iterations to 5 dp to justify 1.18 to 2 dp or show there is a sign change in the interval (1.175, 1.185)	A1	1, 1.16667, 1.17724, 1.17909 1.5, 1.28409, 1.20434, 1.18441, 1.18042 2, 1.55556, 1.30937, 1.21188, 1.18603, 1.18073, 1.17973 Condone minor differences in final figures.
		3	

Question	Answer	Marks	Guidance
9(a)	State or imply the form $\frac{A}{3x+1} + \frac{Bx+C}{x^2+4}$	B1	
	Use a correct method for finding a constant	M1	
	Obtain one of $A = 9$, $B = -4$ and $C = 0$	A1	Allow maximum M1A1 for one or more 'correct' values after B0.
	Obtain a second value	A1	
	Obtain a third value	A1	
		5	
9(b)	Use the correct method to find the first two terms of the expansion of $(3x+1)^{-1}$ or $(x^2+4)^{-1}$ or $\left(1+\frac{1}{4}x^2\right)^{-1}$	M1	E.g. $1-3x$, or $\frac{1}{4}-\frac{1}{16}x^2$ or $1-\frac{1}{4}x^2$.
	Obtain correct unsimplified expansions up to the term in x^3 of each partial fraction	A1 FT	The FT is on A , B and C , e.g. $A(1-3x+9x^2-27x^3) + (Bx+C)\frac{1}{4}(1-\frac{1}{4}x^2)$.
		A1 FT	
	$9-28x+81x^2-\frac{971}{4}x^3$	A1	OE Do not ignore, e.g., multiplication through by 4. Condone if in reverse order.
		4	
9(c)	$ x < \frac{1}{3}$ or $-\frac{1}{3} < x < \frac{1}{3}$	B1	Must be a clear conclusion.
		1	

Question	Answer	Marks	Guidance
10	EITHER: Separate variables correctly	(B1)	Condone missing integrals or missing dx, dy, but not both.
	Obtain term $4\ln(y+3)$	B1	Condone omission of brackets if recovered later.
	Commence integration by parts and reach $axe^{-2x} - \int be^{-2x} dx$	M1	
	Obtain $-\frac{1}{2}xe^{-2x} + \int \frac{1}{2}e^{-2x} dx$	A1	
	Complete integration by parts and obtain $-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} [+c]$	A1	
	Use $x = 0, y = -2$ to evaluate a constant or as limits in a solution containing terms of the form $a \ln(y+3)$, bxe^{-2x} and ce^{-2x}	M1	
	Obtain correct answer in any form	A1	E.g. $4\ln(y+3) = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + \frac{1}{4}$.
	Obtain $y = \exp\left(-\frac{1}{8}xe^{-2x} - \frac{1}{16}e^{-2x} + \frac{1}{16}\right) - 3$	A1)	OE ISW once correct form is seen.
	OR: Alternative Method for Question 10		
	Separate variables correctly	(B1)	Condone missing integrals or missing dx, dy but not both
	Obtain term $\ln(y+3)$	B1	Condone omission of brackets if recovered later
	Commence integration by parts and reach $axe^{-2x} - \int be^{-2x} dx$	M1	
	Obtain $-\frac{1}{8}xe^{-2x} + \int \frac{1}{8}e^{-2x} dx$	A1	

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Question	Answer	Marks	Guidance
10	Complete integration by parts and obtain $-\frac{1}{8}xe^{-2x} - \frac{1}{16}e^{-2x} [+c]$	A1	
	Use $x = 0, y = -2$ to evaluate a constant or as limits in a solution containing terms of the form $a \ln(y + 3)$, bxe^{-2x} and ce^{-2x}	M1	
	Obtain correct answer in any form	A1	E.g. $\ln(y + 3) = -\frac{1}{8}xe^{-2x} - \frac{1}{16}e^{-2x} + \frac{1}{16}$.
	Obtain $y = \exp\left(-\frac{1}{8}xe^{-2x} - \frac{1}{16}e^{-2x} + \frac{1}{16}\right) - 3$	A1)	OE ISW once correct form is seen
	Available marks	8	

Question	Answer	Marks	Guidance
11(a)	Use a correct method to form an equation for line t	M1	Allow one arithmetic error.
	Obtain $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ -2 \\ 3 \end{pmatrix}$	A1	OE Must have $\mathbf{r} = \dots$ or $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \dots$ May use $B \begin{pmatrix} -1 \\ -3 \\ 5 \end{pmatrix}$ in place of A .
		2	

Question	Answer	Marks	Guidance
11(b)	Carry out correct process for evaluating the scalar product of $\begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$ and $their \begin{pmatrix} -5 \\ -2 \\ 3 \end{pmatrix}$	M1	E.g. $(-3 \times -5) + (-4 \times -2) + (5 \times 3)$ or $15 + 8 + 15$ or 38. Allow one slip.
	Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result	M1	
	Obtain answer 29.3° or 0.512°	A1	Allow greater accuracy, 29.33387....
		3	

Question	Answer	Marks	Guidance
11(c)	EITHER: Find $\overrightarrow{AQ}$ (or $\overrightarrow{QA}$) for a general point Q on s , e.g. $\begin{pmatrix} -2-3\mu \\ 2-4\mu \\ 3+5\mu \end{pmatrix}$	(B1)	
	Calculate the scalar product of $\overrightarrow{AQ}$ and the direction vector for s , and equate to 0	*M1	
	Solve and obtain correct solution, e.g. $\mu = -\frac{13}{50}$	A1	
	Carry out method to calculate AQ	DM1	
	Obtain answer 3.69	A1)	Or better, e.g. $\sqrt{\frac{681}{50}}$.
	OR:		
	Find $\overrightarrow{AQ}$ (or $\overrightarrow{QA}$) for a general point Q on s , e.g. $\begin{pmatrix} -2-3\mu \\ 2-4\mu \\ 3+5\mu \end{pmatrix}$	(B1)	
	Use a correct method to express AQ^2 (or AQ) in terms of μ	*M1	
	Obtain a correct equation in any form	A1	E.g. $AQ^2 = (-2-3\mu)^2 + (2-4\mu)^2 + (3+5\mu)^2$.
	Carry out a complete method for finding its minimum	DM1	E.g. $-6(-2-3\mu) - 8(2-4\mu) + 10(3+5\mu) = 0$, $\mu = -\frac{13}{50}$.
	Obtain answer 3.69	A1)	Or better, e.g. $\sqrt{\frac{681}{50}}$.

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Question	Answer	Marks	Guidance
11(c)	OR:		
	Calling (2, 1, 5) P , state $\overrightarrow{PA}$ (or $\overrightarrow{AP}$) in component form, e.g. $\begin{pmatrix} 2 \\ -2 \\ -3 \end{pmatrix}$	(B1)	
	Use a scalar product to find the projection of $\overrightarrow{PA}$ (or $\overrightarrow{AP}$) onto s	*M1	
	Obtain correct answer $\frac{13}{\sqrt{50}}$	A1	OE
	Use Pythagoras to find the perpendicular	DM1	$d = \sqrt{AP^2 - AQ^2} = \sqrt{2^2 + 2^2 + 3^2 - \left(\frac{13}{\sqrt{50}}\right)^2}$
	Obtain answer 3.69	A1)	Or better, e.g. $\sqrt{\frac{681}{50}}$.

Question	Answer	Marks	Guidance
11(c)	OR:		
	Calling (2, 1, 5) P , state $\overrightarrow{PA}$ (or $\overrightarrow{AP}$) in component form, e.g. $\begin{pmatrix} 2 \\ -2 \\ -3 \end{pmatrix}$	(B1)	
	Calculate the vector product of $\overrightarrow{PA}$ and the direction vector for s	*M1	
	Obtain correct answer, e.g. $\begin{pmatrix} -22 \\ -1 \\ -14 \end{pmatrix}$	A1	
	Divide modulus of the product by that of the direction vector	DM1	E.g. $\frac{\sqrt{22^2 + 1^2 + 14^2}}{\sqrt{3^2 + 4^2 + 5^2}}$.
	Obtain answer 3.69	A1)	Or better, e.g. $\sqrt{\frac{681}{50}}$.
	Available marks	5	